CS2006-Operating Systems
Assignment 01

Instructor	Email
Nasreen Akhtar Mahzaib younas M Shahbaz Ayyaz	Nasreen.akhtar@nu.edu.pk mahzaib.younas@nu.edu.pk Shahbaz.ayyaz@nu.edu.pk

Maximum Marks :100

Submission Guidelines:

1- **Deadline:15-02-2024**
2- **Late submissions will not be accepted.**
3- **Ensure your work is original and avoid plagiarism. Plagiarism will result in zero marks for the assignment.**
4- **Only handwritten assignments will be accepted.**

Question 1:

How do modern operating systems handle memory leaks, and what are the potential consequences of unresolved memory leaks in long-running applications?

Question 2:

Describe the concept of process management in an operating system. Discuss process states and the transitions between them.

Question 3:

Discuss the impact of cache memory on memory management strategies, including cache coherence protocols and techniques for optimizing cache utilization in multi-core systems.

Question 4:

Explain the concept of virtualization and its role in modern computing environments. Provide examples of virtualization technologies used in operating systems.

Question 5:

Describe the role of the memory management unit (MMU) in modern computer architectures, and discuss its significance for memory protection and virtual memory implementation.

Question 6:

Explain the role of system utilities in an operating system. Provide examples of common system utilities and their functions.

Question 7:

What role does abstraction play in file-system management, and how does it allow for the manipulation of physical storage properties?

Question 8:

Discuss the benefits of using APIs (Application Programming Interfaces) in software development. Explain how APIs abstract underlying system functionality and provide a standardized interface for building applications.

Question 9:

Provide examples of popular system calls and APIs in modern operating systems, such as POSIX APIs in Unix-like systems and Win32 APIs in Windows. Discuss the functionalities provided by these APIs and their usage in application development.

Question 10:

Describe the operation of a DMA controller and the steps involved in a DMA transfer cycle. Explain how DMA controllers access system memory independently of the CPU and transfer data between memory and I/O devices.

Question 11:

Discuss how NUMA architectures address the scalability and memory access latency challenges of traditional SMP (Symmetric Multiprocessing) systems.

Question 12:

Investigate the impact of register utilization on CPU performance and energy efficiency. How does register allocation affect instruction throughput, pipeline efficiency, and power consumption in modern microarchitectures?